

Homework1

September 14, 2026

1 ORF 405: Regression and Time Series

1.1 Problem 4 (Polynomial regression and cross-validation).

Generate data from:

$$X_i \sim \text{Uniform}[-2, 2], \quad Y_i = 1 + 2X_i - X_i^2 + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2)$$

with $n = 200$.

1.1.1 a. Fit the polynomial regressions of degrees 1, 2, ..., 8 with $\sigma = 1$

```
[56]: import numpy as np
from sklearn.preprocessing import PolynomialFeatures, StandardScaler, OneHotEncoder
from sklearn.linear_model import LinearRegression
from sklearn.pipeline import make_pipeline
import matplotlib.pyplot as plt
from sklearn.model_selection import KFold
import pandas as pd
```

```
[57]: # generate data
np.random.seed(2)
n, sigma = 200, 1.0

X = np.random.uniform(-2, 2, n)
eps = np.random.normal(0, sigma, n)
Y = 1 + 2*X - X**2 + eps
```

```
[58]: models = {}

for d in range(1, 9):
    poly = PolynomialFeatures(degree=d)
    Xd = poly.fit_transform(X.reshape(-1, 1))
    reg = LinearRegression().fit(Xd, Y)
    models[d] = (poly, reg)

x_grid = np.linspace(-2, 2, 200).reshape(-1, 1)
```

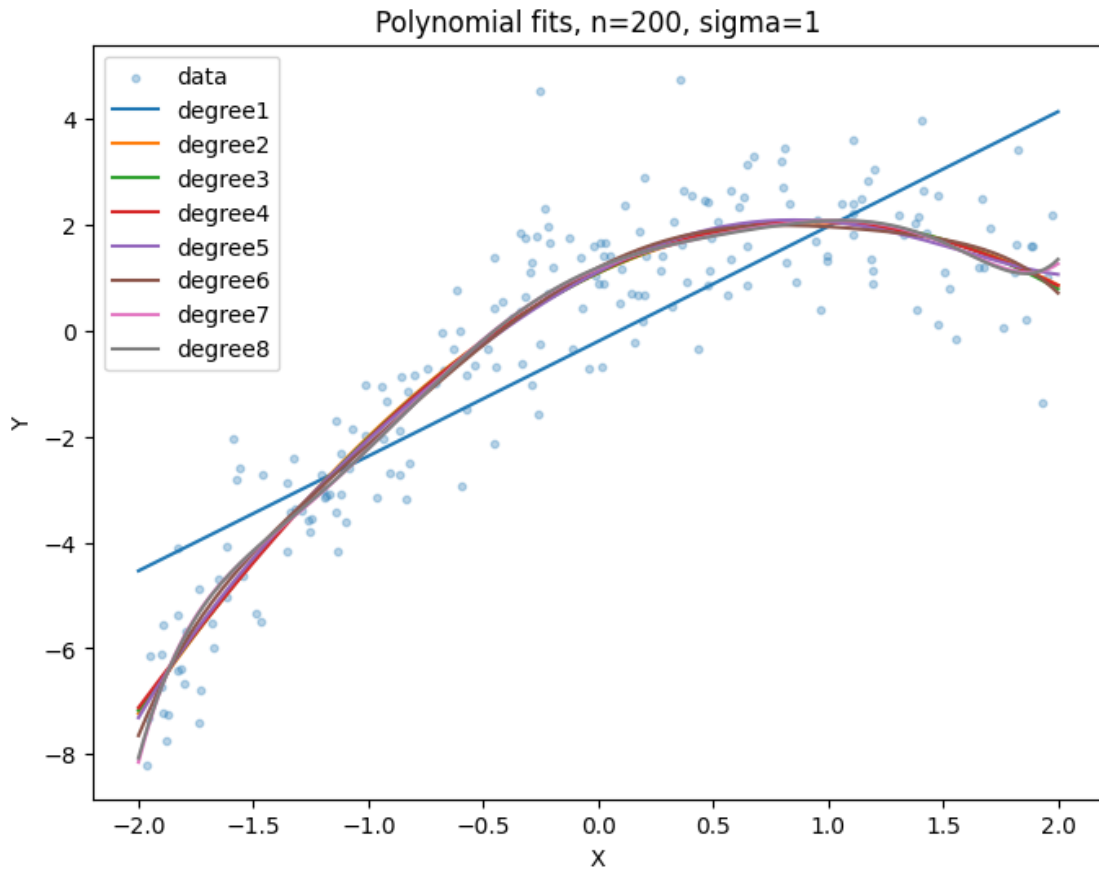
```

plt.figure(figsize=(8, 6))
plt.scatter(X, Y, s=10, alpha=0.3, label="data")

for d in range(1, 9):
    poly, reg = models[d]
    y_pred = reg.predict(poly.transform(x_grid))
    plt.plot(x_grid, y_pred, label = f"degree{d}")

plt.legend()
plt.xlabel("X")
plt.ylabel("Y")
plt.title("Polynomial fits, n=200, sigma=1")
plt.show()

```



1.1.2 b. Use 5-fold cross-validation to choose polynomial degree.

```
[59]: # 5 fold
kf = KFold(n_splits=5, shuffle=True, random_state=1)
cv_errors = {} # parse this

for d in range(1, 9):
    fold_mse = []
    for train_idx, val_idx, in kf.split(X):
        X_train, X_val = X[train_idx], X[val_idx]
        Y_train, Y_val = Y[train_idx], Y[val_idx]

        poly = PolynomialFeatures(degree=d)
        Xd_train = poly.fit_transform(X_train.reshape(-1, 1))
        Xd_val = poly.transform(X_val.reshape(-1, 1))

        reg = LinearRegression().fit(Xd_train, Y_train)
        y_pred = reg.predict(Xd_val)
        fold_mse.append(np.mean((Y_val - y_pred)**2))

    cv_errors[d] = np.mean(fold_mse)

best_d = min(cv_errors, key=cv_errors.get)
print(cv_errors)
print(f"Best Degree: {best_d}")
```

```
{1: np.float64(2.7534719139966244), 2: np.float64(1.1027397002551387), 3:
np.float64(1.1171461276631152), 4: np.float64(1.1281840783694996), 5:
np.float64(1.1400776010516522), 6: np.float64(1.1385052511180125), 7:
np.float64(1.1351498893253353), 8: np.float64(1.1531444283068875)}
Best Degree: 2
```

1.1.3 c. Repeat the experiment for two noise levels: $\sigma = 0.5$ and $\sigma = 2$.

```
[60]: def experiment(sigma, seed=2):
    np.random.seed(seed)
    X = np.random.uniform(-2, 2, n)
    eps = np.random.normal(0, sigma, n)
    Y = 1 + 2*X - X**2 + eps

    kf = KFold(n_splits=5, shuffle=True, random_state=1)
    cv_errors = {}
    for d in range(1, 9):
        fold_mse = []
        for train_idx, val_idx in kf.split(X):
            X_train, X_val = X[train_idx], X[val_idx]
            Y_train, Y_val = Y[train_idx], Y[val_idx]
```

```

poly = PolynomialFeatures(degree=d)
Xd_train = poly.fit_transform(X_train.reshape(-1, 1))
Xd_val = poly.transform(X_val.reshape(-1, 1))

reg = LinearRegression().fit(Xd_train, Y_train)
y_pred = reg.predict(Xd_val)
fold_mse.append(np.mean((Y_val - y_pred)**2))

cv_errors[d] = np.mean(fold_mse)

best_d = min(cv_errors, key=cv_errors.get)
return X, Y, cv_errors, best_d

```

```

[61]: results = {}
for s in [0.5, 2.0]:
    X_s, Y_s, cv_errors_s, best_d_s = experiment(s)
    results[s] = {"X": X_s, "Y": Y_s, "cv_errors": cv_errors_s, "best_d":
    ↪ best_d_s}
    print(f"sigma={s}: best degree = {best_d_s}")
    print(f"cv_errors = {cv_errors_s}")

```

```

sigma=0.5: best degree = 2
cv_errors = {1: np.float64(1.8422133401519099), 2:
np.float64(0.2756849250637847), 3: np.float64(0.2792865319157788), 4:
np.float64(0.28204601959237496), 5: np.float64(0.28501940026291317), 6:
np.float64(0.2846263127795027), 7: np.float64(0.28378747233133383), 8:
np.float64(0.28828610707672225)}
sigma=2.0: best degree = 2
cv_errors = {1: np.float64(6.225563899266594), 2: np.float64(4.410958801020555),
3: np.float64(4.468584510652465), 4: np.float64(4.512736313477999), 5:
np.float64(4.560310404206614), 6: np.float64(4.554021004472046), 7:
np.float64(4.54059955730134), 8: np.float64(4.612577713227567)}

```

1.1.4 Report the cross-validation errors in a table and plot the fitted curve for the selected model.

```

[62]: from IPython.display import display

# combine sigma=1 results (from part b) with sigma=0.5, 2.0 (from part c)
all_results = {
    0.5: results[0.5],
    1.0: {"X": X, "Y": Y, "cv_errors": cv_errors, "best_d": best_d},
    2.0: results[2.0],
}

# --- table of CV errors, degree x sigma ---

```

```

cv_table = pd.DataFrame({s: all_results[s]["cv_errors"] for s in
    ↪sorted(all_results)})
cv_table.index.name = "degree"
cv_table.columns.name = "sigma"
display(cv_table.round(4))

best_degrees = {s: all_results[s]["best_d"] for s in sorted(all_results)}
print("Selected degree by 5-fold CV:", best_degrees)

# --- plot the fitted curve for the CV-selected model, one figure per sigma ---
x_grid = np.linspace(-2, 2, 200).reshape(-1, 1)

for s in sorted(all_results):
    Xs, Ys, best_d_s = all_results[s]["X"], all_results[s]["Y"],
    ↪all_results[s]["best_d"]

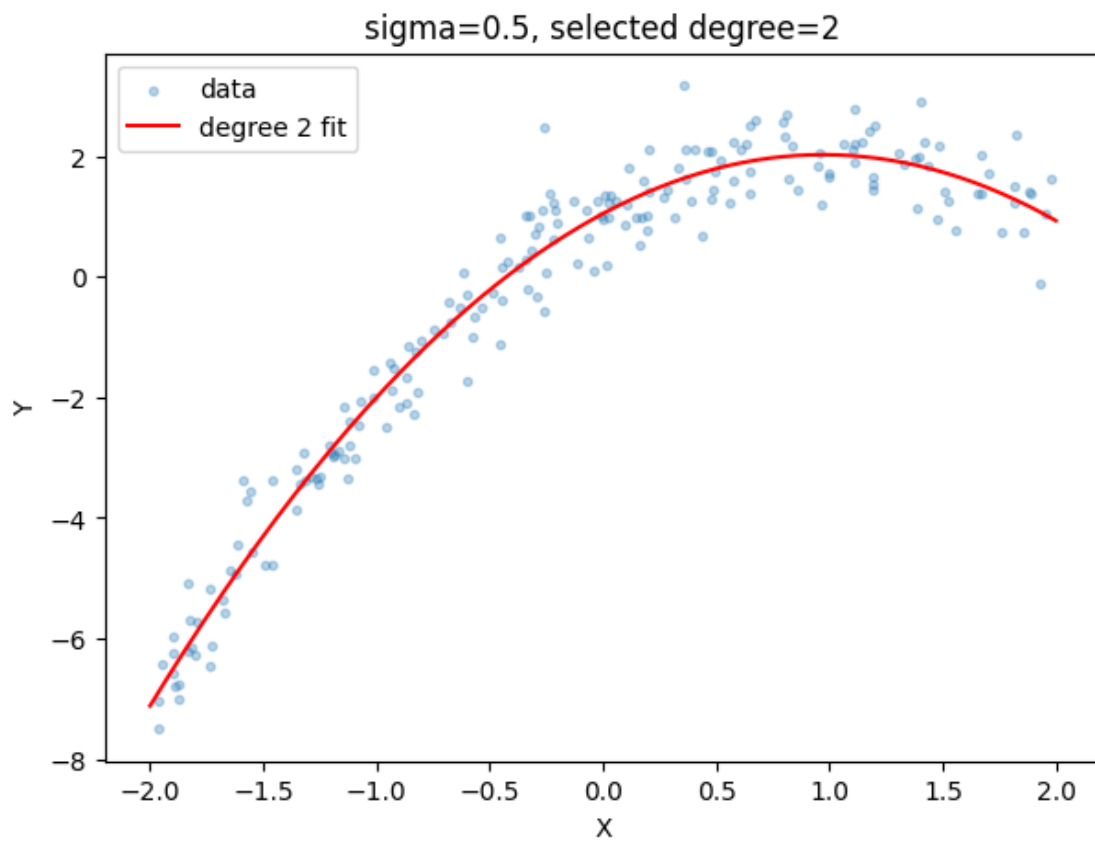
    poly = PolynomialFeatures(degree=best_d_s)
    Xd = poly.fit_transform(Xs.reshape(-1, 1))
    reg = LinearRegression().fit(Xd, Ys)
    y_pred = reg.predict(poly.transform(x_grid))

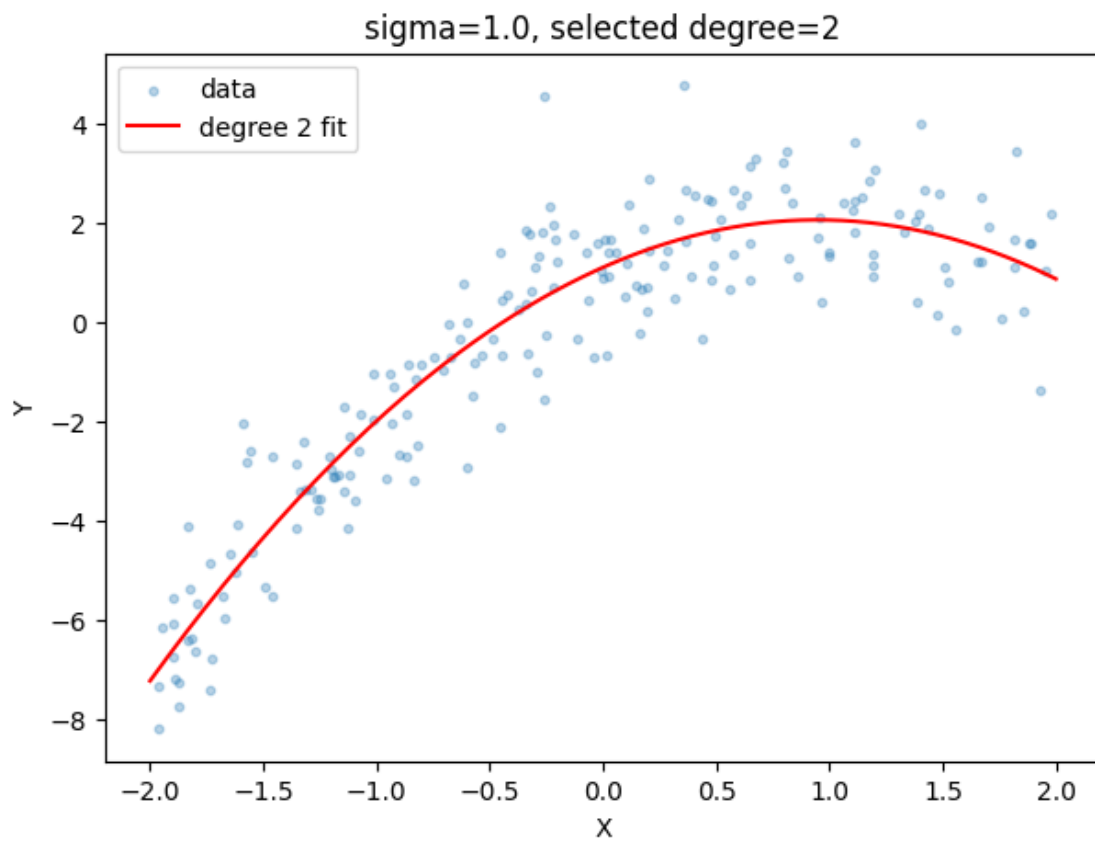
    plt.figure(figsize=(7, 5))
    plt.scatter(Xs, Ys, s=10, alpha=0.3, label="data")
    plt.plot(x_grid, y_pred, color="red", label=f"degree {best_d_s} fit")
    plt.title(f"sigma={s}, selected degree={best_d_s}")
    plt.xlabel("X")
    plt.ylabel("Y")
    plt.legend()
    plt.show()

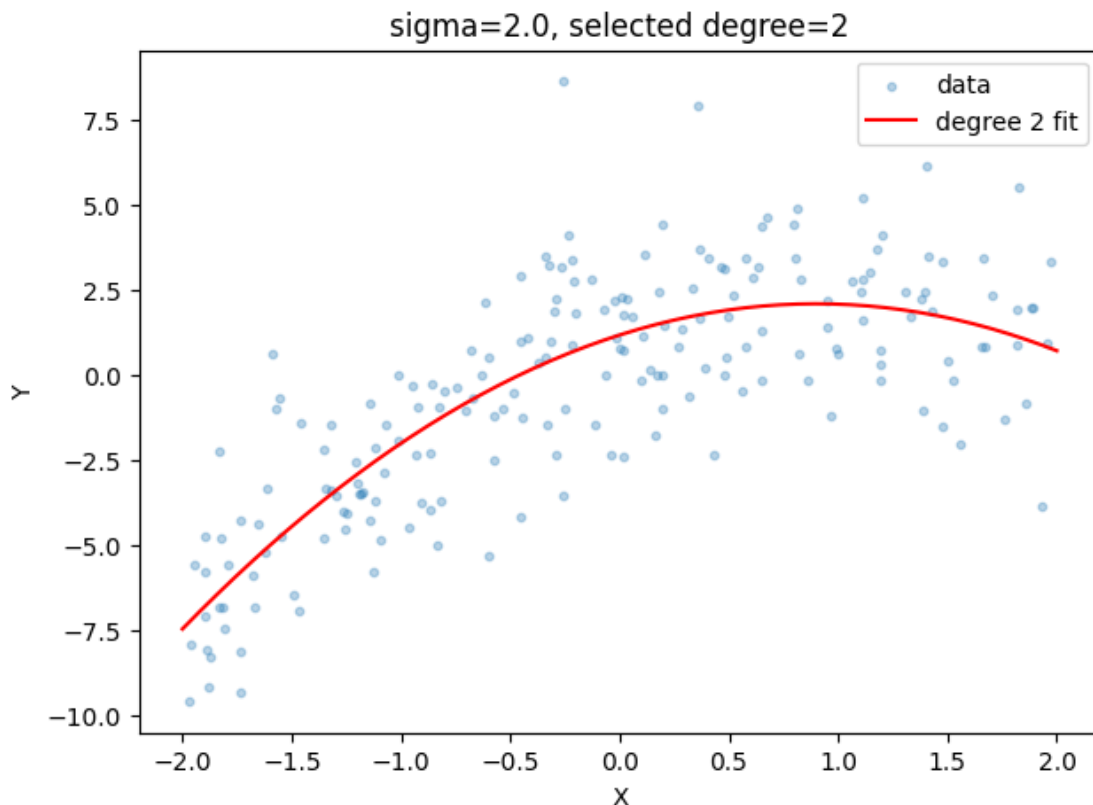
```

sigma	0.5	1.0	2.0
degree			
1	1.8422	2.7535	6.2256
2	0.2757	1.1027	4.4110
3	0.2793	1.1171	4.4686
4	0.2820	1.1282	4.5127
5	0.2850	1.1401	4.5603
6	0.2846	1.1385	4.5540
7	0.2838	1.1351	4.5406
8	0.2883	1.1531	4.6126

Selected degree by 5-fold CV: {0.5: 2, 1.0: 2, 2.0: 2}







1.1.5 e. Discuss how the selected degree changes with the noise level.

5-fold CV selects **degree 2** (the true degree) at all three noise levels ($\sigma = 0.5, 1, 2$): degree 1 is always rejected by a wide margin, while degree 2 beats degree 3 by only $\sim 1.3\%$ at every σ , since the true cubic coefficient is 0 and adding it costs almost no bias. Because the true model lies inside the candidate class for every $d \geq 2$, there's no bias left to trade off against variance as σ grows hence the selected degree stays fixed at 2 regardless of the noise level.

1.2 Problem 5 (Prediction and Housing Prices)

Zillow is an online real estate database company that was founded in 2006. An important task for Zillow is to predict the house price. However, their accuracy has been criticized a lot. According to Fortune, "Zillow has Zestimated the value of 57 percent of U.S. housing stock, but only 65 percent of that could be considered 'accurate' – by its definition, within 10 percent of the actual selling price. And even that accuracy isn't equally distributed". Therefore, Zillow needs your help to build a housing pricing model to improve their accuracy. Download the training and testing data and read the data (training data: 15129 cases, testing data: 6484 cases). Make sure that zip code is treated as factor. Letting τ as a test set, define out-of-sample R^2 as of a prediction method $\{\hat{y}_i^{\text{method}}\}$ as

$$R^2 = 1 - \frac{\sum_{i \in \tau} (y_i - \hat{y}_i)^2}{\sum_{i \in \tau} (y_i - \bar{y}_{\text{train}})^2}$$

where $\bar{y}_{\text{train}} = \text{ave}(\{y_i\}_{i \in \tau_0})$ and τ_0 is the training set.

1.2.1 a. Calculate out-of-sample R^2 using variables “bathrooms”, “bedrooms”, “sqft_living”, and “sqft_lot”.

```
[63]: train_data = pd.read_csv('train.data_.csv')
test_data = pd.read_csv('test.data_.csv')

train_data['zipcode'] = train_data['zipcode'].astype('category')
test_data['zipcode'] = test_data['zipcode'].astype('category')

print(train_data.head())
print(train_data.tail())
print(test_data.head())
print(test_data.tail())
```

	Unnamed: 0	id	date	price	bedrooms	bathrooms	\
0	2	6414100192	20141209T000000	538000	3	2.25	
1	4	2487200875	20141209T000000	604000	4	3.00	
2	5	1954400510	20150218T000000	510000	3	2.00	
3	6	7237550310	20140512T000000	1225000	4	4.50	
4	7	1321400060	20140627T000000	257500	3	2.25	

	sqft_living	sqft_lot	floors	waterfront	...	grade	sqft_above	\
0	2570	7242	2.0	0	...	7	2170	
1	1960	5000	1.0	0	...	7	1050	
2	1680	8080	1.0	0	...	8	1680	
3	5420	101930	1.0	0	...	11	3890	
4	1715	6819	2.0	0	...	7	1715	

	sqft_basement	yr_built	yr_renovated	zipcode	lat	long	\
0	400	1951	1991	98125	47.7210	-122.319	
1	910	1965	0	98136	47.5208	-122.393	
2	0	1987	0	98074	47.6168	-122.045	
3	1530	2001	0	98053	47.6561	-122.005	
4	0	1995	0	98003	47.3097	-122.327	

	sqft_living15	sqft_lot15
0	1690	7639
1	1360	5000
2	1800	7503
3	4760	101930
4	2238	6819

[5 rows x 22 columns]

	Unnamed: 0	id	date	price	bedrooms	bathrooms	\
15124	21609	263000018	20140521T000000	360000	3	2.50	
15125	21610	6600060120	20150223T000000	400000	4	2.50	
15126	21611	1523300141	20140623T000000	402101	2	0.75	
15127	21612	291310100	20150116T000000	400000	3	2.50	
15128	21613	1523300157	20141015T000000	325000	2	0.75	

	sqft_living	sqft_lot	floors	waterfront	...	grade	sqft_above	\
15124	1530	1131	3.0	0	...	8	1530	
15125	2310	5813	2.0	0	...	8	2310	
15126	1020	1350	2.0	0	...	7	1020	
15127	1600	2388	2.0	0	...	8	1600	
15128	1020	1076	2.0	0	...	7	1020	

	sqft_basement	yr_built	yr_renovated	zipcode	lat	long	\
15124	0	2009	0	98103	47.6993	-122.346	
15125	0	2014	0	98146	47.5107	-122.362	
15126	0	2009	0	98144	47.5944	-122.299	
15127	0	2004	0	98027	47.5345	-122.069	
15128	0	2008	0	98144	47.5941	-122.299	

	sqft_living15	sqft_lot15
15124	1530	1509
15125	1830	7200
15126	1020	2007
15127	1410	1287
15128	1020	1357

[5 rows x 22 columns]

	Unnamed: 0	id	date	price	bedrooms	bathrooms	\
0	1	7129300520	20141013T000000	221900	3	1.0	
1	3	5631500400	20150225T000000	180000	2	1.0	
2	11	1736800520	20150403T000000	662500	3	2.5	
3	18	6865200140	20140529T000000	485000	4	1.0	
4	20	7983200060	20150424T000000	230000	3	1.0	

	sqft_living	sqft_lot	floors	waterfront	...	grade	sqft_above	\
0	1180	5650	1.0	0	...	7	1180	
1	770	10000	1.0	0	...	6	770	
2	3560	9796	1.0	0	...	8	1860	
3	1600	4300	1.5	0	...	7	1600	
4	1250	9774	1.0	0	...	7	1250	

	sqft_basement	yr_built	yr_renovated	zipcode	lat	long	\
0	0	1955	0	98178	47.5112	-122.257	
1	0	1933	0	98028	47.7379	-122.233	
2	1700	1965	0	98007	47.6007	-122.145	

3	0	1916	0	98103	47.6648	-122.343
4	0	1969	0	98003	47.3343	-122.306

	sqft_living15	sqft_lot15
0	1340	5650
1	2720	8062
2	2210	8925
3	1610	4300
4	1280	8850

[5 rows x 22 columns]

	Unnamed: 0	id	date	price	bedrooms	bathrooms	\
6479	21596	1972201967	20141031T000000	520000	2	2.25	
6480	21601	249000205	20141015T000000	1537000	5	3.75	
6481	21605	9834201367	20150126T000000	429000	3	2.00	
6482	21606	3448900210	20141014T000000	610685	4	2.50	
6483	21607	7936000429	20150326T000000	1007500	4	3.50	

	sqft_living	sqft_lot	floors	waterfront	...	grade	sqft_above	\
6479	1530	981	3.0	0	...	8	1480	
6480	4470	8088	2.0	0	...	11	4470	
6481	1490	1126	3.0	0	...	8	1490	
6482	2520	6023	2.0	0	...	9	2520	
6483	3510	7200	2.0	0	...	9	2600	

	sqft_basement	yr_built	yr_renovated	zipcode	lat	long	\
6479	50	2006	0	98103	47.6533	-122.346	
6480	0	2008	0	98004	47.6321	-122.200	
6481	0	2014	0	98144	47.5699	-122.288	
6482	0	2014	0	98056	47.5137	-122.167	
6483	910	2009	0	98136	47.5537	-122.398	

	sqft_living15	sqft_lot15
6479	1530	1282
6480	2780	8964
6481	1400	1230
6482	2520	6023
6483	2050	6200

[5 rows x 22 columns]

```
[64]: features_a = ["bathrooms", "bedrooms", "sqft_living", "sqft_lot"]

X_train_a = train_data[features_a]
y_train = train_data["price"]
X_test_a = test_data[features_a]
y_test = test_data["price"]
```

```

reg_a = LinearRegression().fit(X_train_a, y_train)
y_pred_a = reg_a.predict(X_test_a)

y_train_mean = y_train.mean()
r2_oos_a = 1 - np.sum((y_test - y_pred_a)**2) / np.sum((y_test -
    ↪y_train_mean)**2)

print(f"Out-of-sample R^2 (a): {r2_oos_a:.4f}")

```

Out-of-sample R^2 (a): 0.5051

1.2.2 b. Calculate out-of-sample R^2 using the 4 variables above along with interaction terms.

```

[65]: poly_b = PolynomialFeatures(degree=2, interaction_only=True, include_bias=False)

X_train_b = poly_b.fit_transform(train_data[features_a])
X_test_b = poly_b.transform(test_data[features_a])

# standardize: sqft_living*sqft_lot reaches ~5.7e9, which ill-conditions the
    ↪raw fit
reg_b = make_pipeline(StandardScaler(), LinearRegression()).fit(X_train_b,
    ↪y_train)
y_pred_b = reg_b.predict(X_test_b)

r2_oos_b = 1 - np.sum((y_test - y_pred_b)**2) / np.sum((y_test -
    ↪y_train_mean)**2)

print(f"Out-of-sample R^2 (b): {r2_oos_b:.4f}")

```

Out-of-sample R^2 (b): 0.5329

1.2.3 c. Add the factor zipcode to (b) and compute out-of-sample R^2 .

```

[66]: ohe = OneHotEncoder(handle_unknown='ignore', sparse_output=False)
zip_train = ohe.fit_transform(train_data[['zipcode']])
zip_test = ohe.transform(test_data[['zipcode']])

X_train_c = np.hstack([X_train_b, zip_train])
X_test_c = np.hstack([X_test_b, zip_test])

reg_c = make_pipeline(StandardScaler(), LinearRegression()).fit(X_train_c,
    ↪y_train)
y_pred_c = reg_c.predict(X_test_c)

r2_oos_c = 1 - np.sum((y_test - y_pred_c)**2) / np.sum((y_test -
    ↪y_train_mean)**2)

```

```
print(f"Out-of-sample R^2 (c): {r2_oos_c:.4f}")
```

Out-of-sample R² (c): 0.7620

1.2.4 d. Add the following additional variables to (d) (below) and compute out-of-sample R^2 .

$$X_{12} = I(\text{view} = 0)$$

$$X_{13} = L^2$$

$$X_{13+i} = (L - \tau_i)_+^2, \quad i = 1, \dots, 9$$

where τ_i is the $10 \times i$ -th percentile and L is the size of living area ("sqft living").

```
[67]: L_train = train_data["sqft_living"]
      L_test = test_data["sqft_living"]
      taus = np.percentile(L_train, [10 * i for i in range(1, 10)])

      view0_train = (train_data["view"] == 0).astype(int).to_numpy().reshape(-1, 1)
      view0_test = (test_data["view"] == 0).astype(int).to_numpy().reshape(-1, 1)

      L2_train = (L_train ** 2).to_numpy().reshape(-1, 1)
      L2_test = (L_test ** 2).to_numpy().reshape(-1, 1)

      spline_train = np.column_stack([np.maximum(L_train - tau, 0) ** 2 for tau in
      ↪taus])
      spline_test = np.column_stack([np.maximum(L_test - tau, 0) ** 2 for tau in
      ↪taus])

      X_train_d = np.hstack([X_train_c, view0_train, L2_train, spline_train])
      X_test_d = np.hstack([X_test_c, view0_test, L2_test, spline_test])

      reg_d = make_pipeline(StandardScaler(), LinearRegression()).fit(X_train_d,
      ↪y_train)
      y_pred_d = reg_d.predict(X_test_d)

      r2_oos_d = 1 - np.sum((y_test - y_pred_d)**2) / np.sum((y_test -
      ↪y_train_mean)**2)

      print(f"Out-of-sample R^2 (d): {r2_oos_d:.4f}")
```

Out-of-sample R² (d): 0.7805

1.2.5 e. Now without assessing to the test data, using the 5-fold cross-validation and the training data alone, compute the out-of-sample R^2 for the last model.

```
[68]: def build_design(sub_train, sub_val):
    Xb_tr = poly_b.transform(sub_train[features_a])
    Xb_va = poly_b.transform(sub_val[features_a])

    ohe_cv = OneHotEncoder(handle_unknown='ignore', sparse_output=False)
    zip_tr = ohe_cv.fit_transform(sub_train[['zipcode']])
    zip_va = ohe_cv.transform(sub_val[['zipcode']])

    L_tr, L_va = sub_train['sqft_living'], sub_val['sqft_living']
    taus_cv = np.percentile(L_tr, [10 * i for i in range(1, 10)])

    view_tr = (sub_train['view'] == 0).astype(int).to_numpy().reshape(-1, 1)
    view_va = (sub_val['view'] == 0).astype(int).to_numpy().reshape(-1, 1)

    L2_tr = (L_tr ** 2).to_numpy().reshape(-1, 1)
    L2_va = (L_va ** 2).to_numpy().reshape(-1, 1)

    spline_tr = np.column_stack([np.maximum(L_tr - t, 0) ** 2 for t in taus_cv])
    spline_va = np.column_stack([np.maximum(L_va - t, 0) ** 2 for t in taus_cv])

    X_tr = np.hstack([Xb_tr, zip_tr, view_tr, L2_tr, spline_tr])
    X_va = np.hstack([Xb_va, zip_va, view_va, L2_va, spline_va])
    return X_tr, X_va

kf = KFold(n_splits=5, shuffle=True, random_state=1)
sse_total, sst_total = 0.0, 0.0

for tr_idx, va_idx in kf.split(train_data):
    sub_train, sub_val = train_data.iloc[tr_idx], train_data.iloc[va_idx]
    y_tr, y_va = sub_train['price'], sub_val['price']

    X_tr, X_va = build_design(sub_train, sub_val)
    model = make_pipeline(StandardScaler(), LinearRegression()).fit(X_tr, y_tr)
    y_pred_va = model.predict(X_va)

    sse_total += np.sum((y_va - y_pred_va) ** 2)
    sst_total += np.sum((y_va - y_tr.mean()) ** 2)

r2_cv_e = 1 - sse_total / sst_total
print(f"5-fold CV out-of-sample R^2 (e): {r2_cv_e:.4f}")
```

5-fold CV out-of-sample R^2 (e): 0.7810

```
[69]: summary_5 = pd.DataFrame({
    "model": [
```

```

        "(a) bathrooms, bedrooms, sqft_living, sqft_lot",
        "(b) + pairwise interactions",
        "(c) + zipcode factor",
        "(d) + I(view=0), L^2, 9 spline terms",
        "(e) model (d), 5-fold CV on training data alone",
    ],
    "R^2_out": [r2_oos_a, r2_oos_b, r2_oos_c, r2_oos_d, r2_cv_e],
})
summary_5["R^2_out"] = summary_5["R^2_out"].round(4)
summary_5

```

```

[69]:

```

	model	R^2_out
0	(a) bathrooms, bedrooms, sqft_living, sqft_lot	0.5051
1	(b) + pairwise interactions	0.5329
2	(c) + zipcode factor	0.7620
3	(d) + I(view=0), L^2, 9 spline terms	0.7805
4	(e) model (d), 5-fold CV on training data alone	0.7810

```

[71]: plt.figure(figsize=(8, 5))
bars = plt.bar(["(a)", "(b)", "(c)", "(d)", "(e) CV"], summary_5["R^2_out"],
               color=["#9db8d8", "#9db8d8", "#9db8d8", "#9db8d8", "#c0392b"])
for bar, val in zip(bars, summary_5["R^2_out"]):
    plt.text(bar.get_x() + bar.get_width()/2, val + 0.01, f"{val:.4f}",
             ha="center")

plt.ylabel("Out-of-sample $R^2$")
plt.title("Out-of-sample $R^2$ across models")
plt.ylim(0, 0.9)
plt.show()

```

